MOHAMED OUHMMOUCH

Embedded Software Engineer | Low-level C/C++ | RTOS | Hardware bring-up and test

+33 7 67 57 40 28 | ouhmmouch.mohamed@gmail.com | linkedin.com/in/mohamed-ouhmmouch-601b391b8 | Île-de-France, France

PROFILE

Embedded software engineer with experience in low-level C/C++ firmware on ESP32 and STM32 microcontrollers, real-time systems, peripheral drivers and direct hardware interaction. Contributed to a certified product used in the field (Qometer, LinkyReady by Enedis) and built an automated test bench. Interested in firmware that goes from board bring-up to reliable, tested deployment.

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer - Qovoltis Sept 2024 - Present | Arcueil, France

- Low-level embedded firmware on ESP32 for real-time energy measurement systems.
- Contributed to the Qometer, certified LinkyReady by Enedis: reliable real-time communication with Linky meters (network interoperability, dynamic load management).
- Designed and developed an automated test bench; analysed, diagnosed and resolved incidents on embedded systems.

Embedded Systems Engineer (Apprenticeship) - Qovoltis Oct 2023 - Sept 2024 | Arcueil, France

- Developed an ESP32 energy measurement system in C: SPI communication with the STPM34 metering IC and UDP data transmission.
- System integration and validation.

Embedded Software Engineer Intern - Kalray Mar 2023 - Aug 2023 | Montbonnot-Saint-Martin, France

- Developed and evaluated CNN and LSTM models for channel estimation in 4x4 MIMO systems.
- Built Matlab simulation pipelines for dataset generation and analysis; benchmarked accuracy and robustness across scenarios.

Embedded Systems Intern - INTELLCAP SARL Jul 2022 - Aug 2022 | Rabat, Morocco

- Assisted with software integration on embedded platforms; research on emerging technologies (VR).

EDUCATION

Master's in Embedded Software Systems - Université de Bretagne Occidentale 2023 - 2024 | Brest, France

Exchange programme, Electronic Systems - Grenoble INP - Phelma 2022 - 2023 | Grenoble, France

Engineering degree, Embedded Systems - INPT 2020 - 2023 | Rabat, Morocco

Classes Préparatoires MP - Lycée Mohammed VI d'Excellence 2018 - 2020 | Morocco

SKILLS

Embedded: C, C++, ESP32, STM32, FreeRTOS, Linux, Yocto, drivers, HLS
Buses and protocols: SPI, I2C, UART, CAN, Modbus, UDP; reading datasheets
Tools: ESP-IDF, STM32CubeIDE, GDB, CMake, Git, Jira, VS Code
Performance: OpenMP, multithreading, SIMD, cache management, benchmarking, Matlab
Methods: Agile/Scrum, V-Model, test and validation, debugging

LANGUAGES, ACTIVITIES AND INTERESTS

Languages: French, English, Arabic
Activities: Vice-treasurer of the INPT student union; Hardware team member, Track-Alz; organising committee, INPT Olympiads
Interests: Table tennis, badminton, hiking